# pubmed_25425302 ? head

To compare optic nerve head (ONH) morphology between eyes with normal-tension glaucoma (NTG) and primary open-angle glaucoma (POAG). Seventy-eight NTG patients and 78 POAG patients matched according to age and axial length were analyzed. Optic nerve head tilt and torsion were identified from cross-sectional images of optical coherence tomography. The degree of horizontal, vertical, and maximum ONH tilt and torsion was compared between NTG and POAG eyes, and additional comparisons were based on the presence of myopia and the location of the visual field defect. Logistic regression analysis was used to determine the factors related to the degree of ONH torsion. Vertical ($P = 0.610$) and horizontal tilt degree ($P = 0.746$) did not differ between NTG and POAG eyes. However, torsion degree ($P = 0.022$) differed significantly between NTG and POAG eyes. Direction of vertical tilt ($P = 0.040$) and torsion ($P < 0.001$) showed more prevalent superior tilt and torsion in NTG eyes (21.8% and 33.3%, respectively) compared to POAG eyes (10.3% and 10.3%, respectively). Myopic NTG eyes showed greater torsion degree ($P = 0.014$) than nonmyopic NTG eyes, which was not observed in the comparison between myopic and nonmyopic POAG eyes. Only NTG eyes showed a significant difference in the degree of maximum tilt ($P < 0.001$) and torsion ($P < 0.001$) and the direction of vertical tilt ($P < 0.001$) and torsion ($P = 0.010$) by the location of visual field defect. Longer axial length, maximum tilt degree, and diagnosis of NTG were the factors related to the degree of ONH torsion.

Question: Is torsion of the optic nerve head a prominent feature of normal-tension glaucoma?

Answer: Normal-tension glaucoma eyes had a greater ONH torsion compared to POAG eyes with matched axial length. The direction of the ONH tilt and torsion was related to the location of the visual field defect only in NTG eyes.